

DES Pune University, Pune
School of Engineering and Technology
Department of Computer Engineering and Technology
Program: B. Tech in Computer Science and Engineering

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Subject: Python Programming

Assignment No.: 4

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Lab Assignment: 04

Aim:

To write a Python program to perform addition, subtraction, multiplication, and division. The program accepts input from the user and performs the operation accordingly.

Objective:

To understand and implement basic arithmetic operations using Python.

To accept user input for numbers and operators.

To use conditional statements (if-elif-else) to perform different operations based on user choice.

To handle errors such as division by zero and invalid operator input.

Theory:

A calculator program takes two numbers and an operator as input.

Depending on the operator (+, -, *, /), it performs the corresponding arithmetic operation.

Python uses if-elif-else statements to make decisions based on user input.

Division by zero is handled using a conditional check.

Algorithm:

1. Start the program.
2. Input the first number a and the second number b.
3. Input the operator o.
4. If o is +, perform addition.
5. Else if o is -, perform subtraction.
6. Else if o is *, perform multiplication.
7. Else if o is /, check if b is not zero and perform division. Otherwise, display an error.
8. Else, display "Invalid Operator".
9. Stop the program.

Pseudo Code:

```
START  
READ a, b  
READ operator o  
IF o == '+'
```

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```
PRINT a + b
ELSE IF o == '-'
PRINT a - b
ELSE IF o == '*'
PRINT a * b
ELSE IF o == '/'
IF b != 0
PRINT a / b
ELSE
PRINT "Cannot divide by zero"
ELSE
PRINT "Invalid Operator"
END
```

Code:

```
a=float(input("Enter 1st number = "));
b=float(input("Enter 2nd number = "));
o=input("Enter operation you want to perform(+,-,*,/)");
if o=='+':
    print("a + b = ",a+b);
elif o=='-':
    print("a - b = ",a-b);
elif o=='*':
    print("a * b = ",a*b);
elif o=='/':
    if b != 0:
        d= a/b;
        print("a / b = ", round(d,3));
    else:
        print("Cannot divide by zero");
else:
    print("Invalid Operator");
```

Output:

```
Enter 1st number = 10
Enter 2nd number = 5.2
Enter operation you want to perform(+,-,*,/) /
a / b = 1.923
```

Conclusion:

The program successfully performs addition, subtraction, multiplication, and division based on the operator entered by the user. It demonstrates the use of conditional statements, user input handling, and basic arithmetic operations in Python. The program also ensures error handling by checking for division by zero and invalid operators, making it reliable and user-friendly.